

A counterexample to conjecture 00000001096: the VC dimension of degree- $\leq d$ polynomials over $\mathbb{F}_q$ is not $n \cdot d$

Abstract

Conjecture 00000001096 asserts that the VC dimension of polynomial functions of degree $\leq d$ over $\mathbb{F}_q^n$ is exactly $n \cdot d$ when $d < q$. We refute this. For $n = 2$, $d = 2$, $q = 5$ (so $d < q$) the six monomials $1, x, y, x^2, xy, y^2$ of degree ≤ 2 form a 6-dimensional function space on $\mathbb{F}_5^2$, and six explicit points have an invertible evaluation matrix over $\mathbb{F}_5$ (determinant $\equiv 1 \pmod{5}$). Hence all 5^6 labellings of those six points are realisable in the natural generalised $\mathbb{F}_q$ -valued sense, so the VC dimension is at least 6 (in fact exactly 6), whereas the conjecture predicts $n \cdot d = 4$. The verdict is **FALSE**.

1 The notion of VC dimension and the conjecture

The classical Vapnik–Chervonenkis dimension is defined for *binary* function classes: a class $\mathcal{H} \subseteq \{0, 1\}^X$ shatters a finite set $S \subseteq X$ if the restriction map $\mathcal{H} \rightarrow \{0, 1\}^S$ is surjective, and the VC dimension is the largest size of a shattered set. Conjecture 00000001096 concerns polynomial functions with values in the finite field $\mathbb{F}_q$, which are not binary, so it must intend a generalised notion. We use the standard and natural one.

Definition 1 (generalised $\mathbb{F}_q$ -valued VC dimension). *Let X be a set and let $\mathcal{H} \subseteq \mathbb{F}_q^X$ be a class of $\mathbb{F}_q$ -valued functions. A finite set $S = \{s_1, \dots, s_m\} \subseteq X$ is shattered by $\mathcal{H}$ if the evaluation map*

$$\text{ev}_S: \mathcal{H} \rightarrow \mathbb{F}_q^m, \quad f \mapsto (f(s_1), \dots, f(s_m))$$

is surjective; equivalently, if every one of the q^m labellings in $\mathbb{F}_q^S$ is realised by some $f \in \mathcal{H}$. The VC dimension is the supremum of the sizes of shattered sets.

Remark 2. *For binary classes one takes $\mathbb{F}_2 = \{0, 1\}$, and this definition reduces to the classical one, since surjectivity onto $\mathbb{F}_2^m$ means all 2^m binary patterns occur. We emphasise this choice because “VC dimension” is classically stated only for binary classes; the conjecture must be read through a generalised notion such as the above, and under that reading it fails.*

Definition 3 (the function class). *Fix $d \geq 0$ and $q = p^r$ prime power. Write $\mathcal{P}_{n,d} \subseteq \mathbb{F}_q^{\mathbb{F}_q^n}$ for the functions induced by polynomials of total degree $\leq d$ in n variables. These are exactly the $\mathbb{F}_q$ -linear combinations of the monomials $x_1^{a_1} \cdots x_n^{a_n}$ with $a_1 + \cdots + a_n \leq d$, of which there are $\binom{n+d}{n}$.*

The conjecture under consideration is:

$$\dim_{\text{VC}}(\mathcal{P}_{n,d}) = n \cdot d \quad \text{for } d < q. \tag{1}$$

2 The counterexample: $n = 2$, $d = 2$, $q = 5$

Take $n = 2$, $d = 2$ and $q = 5$; the hypothesis $d < q$ holds since $2 < 5$. The monomials of degree ≤ 2 in x, y are

$$m_0 = 1, \quad m_1 = x, \quad m_2 = y, \quad m_3 = x^2, \quad m_4 = xy, \quad m_5 = y^2,$$

so $\dim_{\mathbb{F}_5} \mathcal{P}_{2,2} = 6$ and the conjecture (1) predicts $n \cdot d = 2 \cdot 2 = 4$.

Choose the six points of $\mathbb{F}_5^2$

$$p_0 = (0, 0), \quad p_1 = (0, 1), \quad p_2 = (0, 2), \quad p_3 = (1, 0), \quad p_4 = (1, 1), \quad p_5 = (2, 0).$$

Let $M \in \mathbb{F}_5^{6 \times 6}$ be the evaluation matrix, $M_{ij} = m_j(p_i)$. The entries are integers reduced modulo 5 (we write the representative in $\{0, \dots, 4\}$):

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 2 & 0 & 0 & 4 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 4 & 0 & 0 \end{pmatrix}. \quad (2)$$

3 The determinant is a unit modulo 5

Proposition 4. $\det M \equiv 1 \pmod{5}$. In particular M is invertible over $\mathbb{F}_5$.

Proof. Expand by the Leibniz formula, $\det M = \sum_{\sigma \in S_6} \text{sgn}(\sigma) \prod_{i=0}^5 M_{i,\sigma(i)}$, and reduce each term modulo 5 (using $\text{sgn}(\sigma) \equiv \pm 1$). Summing the $6! = 720$ terms gives $\det M \equiv 1 \pmod{5}$. This finite computation is checked independently by `reproduce.py` (Leibniz sum over all permutations) and formalised in Lean 4 in `lean4/Main.lean` as `det_is_one`. Since 1 is a unit in $\mathbb{F}_5$, the matrix M is invertible. $\square$

For reference, the inverse of M over $\mathbb{F}_5$ is

$$M^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 0 \\ 3 & 0 & 0 & 4 & 0 & 3 \\ 1 & 4 & 0 & 4 & 1 & 0 \\ 3 & 4 & 3 & 0 & 0 & 0 \end{pmatrix}. \quad (3)$$

4 The six points are shattered

Theorem 5. Under the generalised $\mathbb{F}_q$ -valued notion of Definition 1, the set $S = \{p_0, \dots, p_5\}$ is shattered by $\mathcal{P}_{2,2}$. Consequently

$$\dim_{\text{VC}}(\mathcal{P}_{2,2}) \geq 6 > 4 = n \cdot d,$$

and conjecture (1) is false (verdict **FALSE**).

Proof. A labelling is a vector $y = (y_0, \dots, y_5) \in \mathbb{F}_5^6$, recording the desired value y_i at p_i . Since M is invertible (Proposition 4), the coefficient vector

$$c := M^{-1}y \in \mathbb{F}_5^6$$

is well defined for every y , and $Mc = y$. Explicitly, with $c = (c_0, \dots, c_5)$ the coefficients of $c_0 + c_1x + c_2y + c_3x^2 + c_4xy + c_5y^2$, the inverse (3) yields

$$\begin{aligned} c_0 &= y_0, & c_1 &= y_0 + 2y_3 + 2y_5, & c_2 &= y_0 + 2y_1 + 2y_2, \\ c_3 &= 3y_0 + 4y_3 + 3y_5, & c_4 &= y_0 + 4y_1 + 4y_3 + y_4, & c_5 &= 3y_0 + 4y_1 + 3y_2, \end{aligned}$$

all in $\mathbb{F}_5$. Then $Mc = y$ says exactly that the polynomial $f_c = c_0 + c_1x + c_2y + c_3x^2 + c_4xy + c_5y^2 \in \mathcal{P}_{2,2}$ satisfies $f_c(p_i) = y_i$ for all i . Hence the evaluation map $\text{ev}_S: \mathcal{P}_{2,2} \rightarrow \mathbb{F}_5^6$ is surjective, i.e. every one of the $5^6 = 15625$ labellings of S is realised, and S is shattered. $\square$

Remark 6. *The bound is tight for this example. The six monomials $1, x, y, x^2, xy, y^2$ are linearly independent as functions on $\mathbb{F}_5^2$ (their evaluation matrix (2) is invertible), so $\mathcal{P}_{2,2}$ is a 6-dimensional space of functions; evaluation at any $m > 6$ points can never be surjective onto $\mathbb{F}_5^m$ for dimension reasons. Thus $\dim_{\text{VC}}(\mathcal{P}_{2,2}) = 6$ exactly, not merely ≥ 6 . Moreover, even after removing the constant function, the five monomials x, y, x^2, xy, y^2 shatter the five points $p_1, \dots, p_5$: the corresponding 5×5 minor also has determinant $\equiv 1 \pmod{5}$. So the failure is not an artefact of counting the constant monomial.*

Remark 7. *The source of the failure is that the conjecture counts monomials $\binom{n+d}{n}$ for the case $n = 2, d = 2$: $\binom{4}{2} = 6$, not $n \cdot d = 4$. The polynomial “degree” bound constrains the total degree, so polynomials of degree $\leq d$ span the full space of monomials with $a_1 + \dots + a_n \leq d$, whose dimension is $\binom{n+d}{n}$. In the regime $d < q$ distinct monomials do give distinct functions on $\mathbb{F}_q^n$, so the VC dimension (in the generalised $\mathbb{F}_q$ -valued sense) equals this dimension whenever the corresponding evaluation matrix can be made invertible. For $n = 2, d = 2, q = 5$ that value is 6, contradicting $n \cdot d = 4$.*

Reproducibility

`reproduce.py` (standard library only) builds the matrix (2), computes $\det M \bmod 5 = 1$, verifies $\text{rank}_{\mathbb{F}_5} M = 6$, inverts M over $\mathbb{F}_5$, and checks all $5^6 = 15625$ labellings explicitly; it prints `PASS` and exits 0. The Lean 4 development in `lean4/` uses only core Lean plus `Std` (no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`) and formalises the determinant identity `det_is_one`, the exhaustive component-wise shattering check `shatters_components` over all 5^6 labellings, the function-level statement `eval_map_bijective`, and the assembled `conjecture_00000001096_false`.